Question 1 (Cloud Introduction)

What is the defining characteristic of **on-demand self-service** in cloud computing?

Answer Choices:

A. The service provider controls all resource changes.

B. Users can provision computing resources automatically without human interaction from the provider.

C. Resources are limited to a single geographic region.

D. The pricing model is fixed, regardless of usage.

Question 2 (XaaS Definitions)

Which cloud service model provides the consumer with the ability to deploy and run arbitrary software, including operating systems and applications, but they manage these components?

Answer Choices:

A. Software as a Service (SaaS)

B. Platform as a Service (PaaS)

C. Infrastructure as a Service (IaaS)

D. Function as a Service (FaaS)

Question 3 (Virtualization Basics)

What software layer is used to create and manage virtual machines (VMs) and allow them to run multiple operating systems concurrently on a single physical host?

Answer Choices:

A. Docker Engine

B. Hypervisor

C. Operating System Kernel

D. Load Balancer

Question 4 (Cloud Deployment)

Which deployment model involves cloud infrastructure provisioned for exclusive use by a single organization comprising multiple consumers?

Answer Choices:

A. Public Cloud

B. Private Cloud

C. Hybrid Cloud

D. Community Cloud

Question 5 (Containers)

What is the core difference between a **Virtual Machine (VM)** and a **Container**?

Answer Choices:

A. VMs are less secure than containers.

B. Containers virtualize the hardware; VMs virtualize the OS kernel.

C. VMs include their own isolated guest OS; Containers share the host OS kernel.

D. Containers require a specialized hypervisor; VMs do not.

Question 6 (Docker Basics)

What is the name of the file used by Docker to define the steps required to build a container image?

Answer Choices:

A. Docker Compose

B. Image Manifest

C. Dockerfile

D. YAML file

Question 7 (Kubernetes Components)

In a Kubernetes cluster, what is the smallest deployable unit that holds one or more containers and shared storage?

Answer Choices:

A. Node

B. Cluster

C. Service

D. Pod

Question 8 (SaaS Example)

Which of the following is a classic example of a **Software as a Service (SaaS)** offering?

Answer Choices:

A. Amazon EC2

B. Google Compute Engine (GCE)

C. Microsoft Azure Virtual Machines

D. Microsoft 365 (Office online)

Question 9 (Hypervisor Types)

Which type of hypervisor runs directly on the host machine's hardware, often referred to as "bare-metal"?

Answer Choices:

A. Type 2 Hypervisor

B. Container Hypervisor

C. Type 1 Hypervisor

D. Operating System Virtualization

Question 10 (Virtualization Pitfall)

Which term describes a drawback of virtualization where too many virtual machines are placed on a single physical host, potentially degrading performance for all of them?

Answer Choices:

A. VM Migration

B. Vendor Lock-in

C. VM Sprawl

D. Over-subscription / Over-commitment

Question 11 (Cloud Benefits)

Which characteristic of cloud computing allows providers to dynamically allocate or reallocate resources as consumer demand fluctuates?

Answer Choices:

A. Measured Service

B. Resource Pooling

C. Elasticity / Rapid Elasticity

D. Broad Network Access

Question 12 (Virtualization Need)

What is the primary motivation for enterprises to use virtualization techniques like VM provisioning?

Answer Choices:

A. To reduce the complexity of application development.

B. To maximize the utilization of underlying physical server hardware.

C. To completely eliminate all security risks.

D. To strictly adhere to proprietary software licenses.

Question 13 (Hypervisor)

What is the primary role of the **Hypervisor** in a virtualized environment?

Answer Choices:

A. To provide persistent storage for containers.

B. To manage the physical network connection.

C. To isolate virtual machines and manage their access to physical hardware resources.

D. To decrypt data sent over the network.

Question 14 (IaaS Components)

Which of the following is a core component provided by an **Infrastructure as a Service (IaaS)** vendor?

Answer Choices:

A. Pre-installed applications (e.g., ERP software)

B. Middleware and runtime environment (e.g., Java runtime)

C. Virtual machines, virtual networks, and virtual storage

D. Managed DNS and CDN services

Question 15 (PaaS)

In a **Platform as a Service (PaaS)** model, which component is typically managed by the **consumer/user**?

Answer Choices:

A. Operating System

B. Application Runtime Environment

C. Application Code and Data

D. Storage Infrastructure

Question 16 (VM Provisioning)

What process is essential for automatically cloning a virtual machine (VM) image and applying initial network settings and user accounts before the VM is started for the first time?

Answer Choices:

A. VM Migration

B. Live Debugging

C. Virtual Machine Provisioning / Instantiation

D. Image repository cleanup

Question 17 (Storage as a Service)

When discussing **Storage as a Service (StaaS)**, what is the benefit of implementing **RAID (Redundant Array of Independent Disks)** at the storage backend?

Answer Choices:

A. It provides automatic OS patching.

B. It increases network bandwidth to the storage.

C. It improves data availability and fault tolerance.

D. It performs real-time code compilation.

Question 18 (Cluster Architecture)

What is the primary purpose of a **High-Availability (HA) Cluster** in a cloud or virtualized environment?

Answer Choices:

A. To aggregate compute power for highly parallel tasks.

B. To ensure continuous uptime by automatically failing over services to redundant nodes upon failure.

C. To provide centralized logging and monitoring.

D. To simplify virtual machine creation.

Question 19 (OpenStack)

What is the primary function of the **Horizon Dashboard** interface in an OpenStack cloud environment?

Answer Choices:

A. To manage the physical hardware servers.

B. To provide a web-based graphical user interface (GUI) for users and administrators to interact with cloud resources.

C. To monitor network traffic at Layer 3.

D. To run containerized applications.

Question 20 (OpenStack)

In OpenStack terminology, what is a **Flavor**?

Answer Choices:

A. A template for a network firewall rule.

B. A definition of the compute, memory, and storage capacity available to a virtual machine instance.

C. The name of the OpenStack scheduler service.

D. A unique identifier for a container image.

Question 21 (Docker Networking)

When using the Docker command docker run -p 8080:80 [image], what does the -p 8080:80 flag accomplish?

Answer Choices:

A. It sets the internal container port to 8080.

B. It maps host port 80 to container port 8080.

C. It maps host port 8080 to internal container port 80 (Port Mapping).

D. It changes the default IP address of the container.

Question 22 (Container Orchestration)

What primary issue does **Container Orchestration** (like Kubernetes) solve in a large-scale deployment?

Answer Choices:

A. Eliminating the need for firewalls.

B. Automating the deployment, scaling, healing, and management of multiple containers across a cluster.

C. Reducing the size of individual container images.

D. Providing a dedicated IDE for developers.

Question 23 (Kubernetes Components)

Which object ensures that a specified number of identical Pod replicas are running at any given time, automatically replacing failed Pods?

Answer Choices:

A. Service

B. Node

C. ReplicaSet

D. Ingress

Question 24 (Web Services Comparison)

When comparing various web services, which metric is most directly related to the cloud pricing model known as **Measured Service**?

Answer Choices:

A. User interface complexity

B. Application programming language

C. Usage consumption (e.g., per hour, per GB, per request)

D. Geographic location of the data center

Question 25 (Virtual Machine Migration)

What is the primary benefit of **Live Virtual Machine Migration**?

Answer Choices:

A. It completely eliminates the need for backups.

B. It allows VMs to be moved between physical hosts without any significant service interruption.

C. It makes the VM instantly smaller.

D. It converts a Type 2 Hypervisor to a Type 1 Hypervisor.

Question 26 (Cloud Challenges)

What challenge in the cloud environment is addressed by using a **Hybrid Cloud** model?

Answer Choices:

A. Eliminating all latency for local applications.

B. Addressing regulatory and security concerns by keeping sensitive data/workloads in a Private Cloud while utilizing Public Cloud for scale.

C. Reducing the total number of physical servers to zero.

D. Standardizing all operating systems to Linux.

Question 27 (Organizational Scenarios)

A large software company decides to use an external provider's platform (e.g., AWS Elastic Beanstalk or Azure App Service) to host and manage all its development and testing environments. Which model is this organization primarily utilizing?

Answer Choices:

A. SaaS

B. IaaS

C. PaaS

D. Daas (Desktop as a Service)

Question 28 (Container Deployment)

Which Kubernetes object is primarily used to expose an application running in a set of Pods to the network, either internally or externally, often via load balancing?

Answer Choices:

A. Deployment

B. DaemonSet

C. Service

D. ConfigMap

Question 29 (Operating System Virtualization)

Operating System Virtualization (like containers) is differentiated from traditional VM virtualization by:

Answer Choices:

A. It requires specialized hardware assist technology.

B. It provides full hardware emulation for the guest OS.

C. It shares the host operating system kernel and isolates processes at the OS level.

D. It must use a Type 2 Hypervisor.

Question 30 (Docker Compose)

What is the purpose of the **Docker Compose** tool?

Answer Choices:

A. To create a single container image from a Dockerfile.

B. To manage and run multi-container Docker applications defined in a YAML file.

C. To publish an image to Docker Hub.

D. To monitor the performance of running containers.

Question 31 (Cloud Pricing)

In the context of AWS, Azure, and GCP, which resource is typically charged based on the amount of data transferred **out** of the cloud provider's network (egress) but is often free for data transferred **into** the network (ingress)?

Answer Choices:

A. Virtual CPU (vCPU) utilization

B. Network Traffic

C. Persistent Disk Storage

D. Managed Database Queries

Question 32 (VM Provisioning Workflow)

In a detailed VM provisioning workflow, what is the correct sequence for creating a VM instance after a user request?

Answer Choices:

A. Allocate IP $\rightarrow$ Select Image $\rightarrow$ Select Flavor $\rightarrow$ Attach Storage

B. Select Flavor $\rightarrow$ Select Image $\rightarrow$ Allocate Resources (Compute/Storage) $\rightarrow$ Network/IP Assignment

C. Network/IP Assignment $\rightarrow$ Launch Instance $\rightarrow$ Select Flavor $\rightarrow$ Select Image

D. Select Image $\rightarrow$ Check IP $\rightarrow$ Launch Instance $\rightarrow$ Attach Storage

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Question 33 (Type 1 Hypervisor Challenge)

A significant challenge in the design and implementation of a Type 1 (bare-metal) Hypervisor is:

Answer Choices:

A. Its reliance on a single operating system kernel.

B. The difficulty in developing and managing hardware device drivers for the guest operating systems.

C. Its inability to support Linux operating systems.

D. Its dependency on high-latency networking.

Question 34 (Kubernetes Abstraction)

Which Kubernetes resource serves as a key abstraction layer, representing the desired state (e.g., image, replicas, update strategy) of a set of Pods, enabling rolling updates and rollbacks?

Answer Choices:

A. ReplicaSet

B. Deployment

C. Node

D. Secret

Question 35 (IaaS vs. PaaS Control)

If a user needs to install a custom, non-standard library and tune the kernel of the underlying operating system for an application, which cloud service model provides the necessary level of control?

Answer Choices:

A. SaaS (Software as a Service)

B. PaaS (Platform as a Service)

C. IaaS (Infrastructure as a Service)

D. StaaS (Storage as a Service)

Question 36 (Virtualization Pitfall - VM Sprawl)

VM Sprawl is a management challenge that primarily results in:

Answer Choices:

A. Underutilized network bandwidth.

B. Excessive and unmanaged virtual machines consuming resources, leading to increased licensing costs and security risks.

C. The complete failure of the host machine.

D. Guaranteed resource contention due to over-commitment.

Question 37 (Cloud Vendor Characteristics)

A defining characteristic of a major **Public Cloud Vendor** (like AWS, Azure, or GCP) that enables global scale is:

Answer Choices:

A. Centralized, single data center architecture.

B. Ownership and management of a global, highly redundant fiber optic backbone network.

C. Exclusive use of a Type 2 Hypervisor.

D. Mandating the use of a proprietary operating system.

Question 38 (Container Image Creation)

In a Dockerfile for a Python application, the instruction CMD ["python", "app.py"] defines:

Answer Choices:

A. A command that must be executed during the image build process.

B. The default command that is executed when a container starts from the image.

C. An environment variable named "CMD".

D. A command to be run only if the image build fails.

Question 39 (Cloud Pricing Model)

The cloud computing paradigm generally supports a pricing model known as **Utility Computing**. Which of the following best describes this model?

Answer Choices:

A. Paying a single, fixed fee for unlimited usage annually.

B. Paying only for the exact resources (compute, storage, network) consumed, measured per second, hour, or byte.

C. Paying based on the geographic distance to the data center.

D. Paying a flat rate for access to all available services.

Question 40 (Security Groups)

In cloud environments like OpenStack or AWS, a **Security Group** acts as a virtual firewall at which point?

Answer Choices:

A. At the physical host network switch.

B. At the Operating System kernel level of the VM.

C. At the instance (VM) interface level, controlling inbound and outbound traffic.

D. At the Hypervisor management interface.